ment, which became very popular among our customers in global manufacturing companies due to its transportability, repeatability, and stability (TRS). Next came the 3070 Series 2 launched in 1989, Series 3 in 1998, Series 4 and 5 followed, and the latest generation, the 3070 Series 6, was introduced in 2019. We continue to offer compatibility with our old products, allowing our customers to continue using our ICT system as we offer upgrades from the old series to the new 3070 Series 6. Instead of obsoleting the existing manufacturing tests, we are innovating and translating them to test new technologies that our customers are putting in their PCBs.

What is an 'end-to-end electronics manufacturing' test?

For the past two decades, electronics manufacturing tests have predominantly focused on compliance at the end of the line, leaving a significant unexplored space at the beginning. Despite this hardware-centric approach, which involves early-stage component detection and subsequent electrical measurements, a gap exists. Now, with the implementation of end-to-end manufacturing analysis, specifically using of pathway manufacturing analysis (PMA) software, a solution has emerged.

PMA allows us to correlate failures at the end of the line with issues at the beginning, bridging the previously overlooked gap. By detecting early signs of component failure or a sudden increase in failures, the software enables immediate notification to the beginning of the line, preventing the completion of an entire defective production run for the day, saving valuable time and resources. Additionally, predictive analysis within the software assesses whether a component is degrading in value, considering factors like system-related issues or oven temperature. PMA informs operators of anomalies that require prompt attention.

Where does your PMA software get all its data from, for it to analyse?

For PMA deployment, the customer needs to provide a small server (with the configuration of a desktop/laptop), called a data collection agent (DCA), which can communicate with the PMA server and collect logs from all production machines. For Keysight ICT machines, ICT software is pre-installed with the utility that sends test logs to the DCA. The utility needs to be configured so that it starts sending data to the DCA. Once the DCA receives the logs, it sends them to the PMA server for analysis. For non-Keysight ICTs and FCTs, all test